ISHRAQ TASHDID

tashdid.ishraq@gmail.com — ishraq.tashdid@ucf.edu — +1 (689) 286-7719
Orlando, FL 32817 — LinkedIn — GitHub — Google Scholar

Research Interests

Automated vulnerability reasoning and threat modeling (CWE/CVE/CAPEC); LLM-assisted security analysis across hardware–firmware–software boundaries; formal and property-based security verification; chiplet and SiP authentication; secure SoC design and supply chain integrity.

Education

University of Central Florida (UCF) *Jan. 2024–Present*
Ph.D. in Computer Engineering (Joint PhD/MS) GPA: **3.96**/4.00
Advisor: Dr. Sazadur Rahman, Department of Electrical & Computer Engineering
Trustee's Doctoral Fellow (4-year, $25,000/year) **Future Faculty Laureate**, UCF ECE

Bangladesh University of Engineering and Technology (BUET) *Feb. 2017–Jun. 2022*
B.Sc. in Electrical and Electronic Engineering

Research Experience

Graduate Research Assistant — University of Central Florida *Jan. 2024–Present*
Hardware Security & Trustworthy Computing Lab, advised by Dr. Sazadur Rahman

- **LLM-driven threat modeling & automated vulnerability reasoning (DAC 2026):** Built *ATLAS*, a framework that ingests CWE/CVE/CAPEC vulnerability databases, auto-generates structured threat models for any SoC asset, and translates them into formal security properties verified by JasperGold. Detected 39/48 known CWEs across three HACK@DAC benchmarks (>82% correct property generation). Directly applicable to firmware and software vulnerability workflows that reason over the same threat taxonomies.
- **Automated security property generation & cross-abstraction analysis:** Developed pipelines combining static analysis, AST-level structural reasoning, execution traces, and formal verification feedback to generate precise security assertions with minimal human effort, reducing reliance on manually crafted security rules at every abstraction layer from RTL through system firmware.
- **Chiplet authentication via PUF and MPC (HOST 2026):** Designed *InterPUF*, a distributed authentication framework for reconfigurable System-in-Package (SiP) interposers. Embedded a differential delay PUF in the interconnect fabric and secured evaluation using multi-party computation (MPC), ensuring raw PUF signatures never leave the device. Achieved <0.23% area and <0.072% power overhead; ML modeling attacks converged at ∼47% accuracy (random-guess level).
- **Adaptive logic obfuscation for SoC IP protection (ICCD 2025):** Developed *ECOLogic*, a framework enabling circular, obfuscated, and adaptive logic via eFPGA-augmented SoCs. The approach allows post-fabrication logic reconfiguration to resist reverse engineering and hardware Trojan insertion while maintaining functional correctness and low overhead.
- **Macro placement automation (MLCAD 2025 — Best Paper Nominee):** Designed human-inspired reinforcement learning frameworks for chip floorplanning to improve post-route routability and reliability in heterogeneous SoCs.
- **Scalable chiplet trust architectures:** Developed *SAFE-SiP* (GLSVLSI 2025) and *AuthenTree* (PAINE 2025), two complementary MPC-based distributed trust frameworks addressing authentication scalability in chiplet-based heterogeneous integration. AuthenTree achieves logarithmic communication complexity in the number of chiplets, enabling practical deployment in large-scale SiP designs.

Industry Experience

XPU Security Research Intern — Intel Corporation, Folsom, CA *May 2026–Present*

- Conducting security research on next-generation XPU (accelerator) architectures with Intel's AI and hardware security research team.

Hardware Verification Engineer — XCelerium, Irvine, CA *Jan. 2022–Dec. 2023*

- Verified custom **RISC-V** processor implementations at the ISA and **system level** using SystemVerilog, UVM, and cocotb, developing deep familiarity with the hardware–software boundary where firmware executes and privilege/mode transitions occur.
- Verified **AXI memory controllers** and **APB-AXI bridges**, including CDC edge cases and assertion-based checks on bus-level access control — the same protocol surfaces targeted by firmware and software security assessments.
- Built **Python-driven automation frameworks** for coverage convergence, testbench orchestration, and results analysis, accelerating sign-off across multiple projects.

Selected Publications

- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, *"ATLAS: AI-Assisted Threat-to-Assertion Learning for System-on-Chip Security Verification,"* **DAC 2026** (63rd Design Automation Conference), Long Beach, CA, USA. *Accepted.*
- **Ishraq Tashdid**, *"RoutePUF: A Routing-Based Physically Unclonable Function for Secure Interposer Authentication in Chiplet Systems,"* **GOMACTech 2026**. *Accepted.*
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, *"InterPUF: Distributed Authentication via Physically Unclonable Functions and Multi-party Computation for Reconfigurable Interposers,"* **HOST 2026** (IEEE Int'l Symposium on Hardware Oriented Security and Trust), Washington, D.C., USA. *Accepted.*
- **Ishraq Tashdid**, Dewan Saiham, Nafisa Anjum, Tasnuva Farheen, Sazadur Rahman, *"ECOLogic: Enabling Circular, Obfuscated, and Adaptive Logic via eFPGA-Augmented SoCs,"* **ICCD 2025** (IEEE Int'l Conference on Computer Design), Dallas, TX, USA.
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, *"SAFE-SiP: Secure Authentication Framework for System-in-Package Using Multi-party Computation,"* **GLSVLSI 2025** (ACM Great Lakes Symposium on VLSI), New Orleans, LA, USA.
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, *"AuthenTree: A Scalable MPC-Based Distributed Trust Architecture for Chiplet-based Heterogeneous Systems,"* **PAINE 2025**, Denver, CO, USA.
- **Ishraq Tashdid**, Valentina Terry, Jordan Merkel, Tasnuva Farheen, Sazadur Rahman, *"BeyondPPA: Human-Inspired Reinforcement Learning for Post-Route Reliability-Aware Macro Placement,"* **MLCAD 2025** (ACM/IEEE Workshop on Machine Learning for CAD), Santa Cruz, CA, USA. **Best Paper Nominee.**
- **Book Chapter (invited):** *LLMs for SoC Design and Security* — with collaborators at the University of Florida, *in preparation*, 2025–2026.

Invited Talks

Intel AI Forum — Intel Corporation, Folsom, CA *Feb. 2026*
Invited talk on *ATLAS*: LLM-driven threat modeling and formal security verification for SoC security, presented to Intel's AI research community.

4th Annual Florida Semiconductor Summit (FSI) — Rosen Shingle Creek, Orlando, FL *Feb. 2026*
Invited presentation on hardware security and AI-assisted verification for Florida's semiconductor manufacturing ecosystem. Theme: *Semiconductor Manufacturing in Florida: Power. Progress. Possibilities.*

Teaching & Mentoring

Graduate Teaching Assistant — UCF ECE *Spring 2024; Summer & Fall 2025*

- Courses: Engineering Analysis and Computation; Linear Systems II; Electronics II Lab; Digital Systems Lab. Responsibilities included office hours, grading, and lab instruction.

Undergraduate Research Mentor *Aug. 2024–Present*

- Mentored two undergraduate researchers in the UCF hardware security lab. Both students secured summer 2025 research internships at **AMD** and **Northrop Grumman**, respectively, and co-authored the MLCAD 2025 Best Paper Nominee.

Professional Service

External Reviewer, IEEE International Conference on Computer Design (ICCD 2025)
External Reviewer, IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2026)

Honors, Awards & Fellowships

- **UCF ECE Outstanding Student Researcher Award** (May 2026)
- **DAC Young Fellow**, 63rd Design Automation Conference (DAC 2026)
- **UCF Trustees Doctoral Fellowship** (2024–2028) — 4-year fellowship ($25,000/year) awarded to newly enrolled doctoral students with outstanding academic and research records.
- **Future Faculty Laureate (FFL)** — UCF ECE Department (Fall 2024–Present). Competitive program identifying Ph.D. students with strong potential for academic faculty careers.
- **Best Paper Award Nominee** — ACM/IEEE MLCAD 2025
- **NSF Travel Grant** — IEEE HOST 2025

Technical Skills

Security & Analysis:	Threat modeling (CWE/CVE/CAPEC), LLM-assisted vulnerability reasoning, assertion-based verification, formal verification (JasperGold), static analysis, hardware IP protection
Programming:	Python, SystemVerilog, Verilog, C, MATLAB, Assembly (RISC-V)
Verification:	UVM, cocotb, pytest, ABV; RISC-V ISA & system-level compliance; AXI/APB bus protocols
EDA & Simulation:	Cadence Innovus / Genus / JasperGold, ModelSim, Quartus Prime, Yosys, PyVerilog
ML / AI:	PyTorch, reinforcement learning (PPO/DDPG), LLM prompting & fine-tuning, RAG pipelines